GSVA Single-Sample Enrichment

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Introduction

GSVA (Gene Set Variation Analysis) transforms a gene-by-sample expression matrix into a pathway-by-sample matrix by assigning each sample a per-pathway enrichment score. The transformation enables pathways to be treated as features in any downstream analysis — Cox regression for survival, clustering, classification, or differential analysis at the pathway level. Where over-representation analysis (GO, KEGG over-representation) tells you which pathways are enriched in a hit list, GSVA tells you how each individual sample stands relative to a panel of pathways.

Prerequisites

A working understanding of gene-set enrichment, log-normalised expression matrices, and the difference between sample-level and contrast-level enrichment.

Theory

For each sample, GSVA computes an empirical CDF of expression values across genes (rank-based) and derives a Kolmogorov-Smirnov-like statistic for each gene set. The result is a per-sample, per-pathway score, typically on a roughly $[-1, 1]$ scale. ssGSEA (single-sample GSEA) is a related method differing primarily in the weighting scheme; both are implemented in the `GSVA` package.

The transformed pathway-by-sample matrix can be analysed with any standard tool: limma for differential pathway expression, Cox regression for survival, hierarchical clustering, supervised classification.

Assumptions

Expression is consistently normalised across samples; gene sets are accurately curated and relevant to the biology; scores are interpretable as relative within a study but not on an absolute scale across studies.

R Implementation

```r
library(GSVA)

set.seed(2026)
n_genes <- 500; n_samples <- 20
expr <- matrix(rnorm(n_genes * n_samples), n_genes, n_samples)
rownames(expr) <- paste0("g", 1:n_genes)

pathways <- list(
  path_A = paste0("g", sample(1:n_genes, 30)),
  path_B = paste0("g", sample(1:n_genes, 40)),
  path_C = paste0("g", sample(1:n_genes, 25))
)

gsva_scores <- gsva(expr, pathways, method = "gsva", kcdf = "Gaussian",
                    verbose = FALSE)
```

```r
dim(gsva_scores)        # pathways x samples
round(gsva_scores[, 1:4], 2)
```

Output & Results

`gsva()` returns a pathway-by-sample matrix of enrichment scores. Each column is one sample's pathway profile; high scores indicate the pathway's genes are coordinately up-regulated in that sample relative to others, low scores indicate the opposite.

Interpretation

A reporting sentence: "GSVA pathway scores (Hallmark v7.5 gene sets, log-CPM input, Gaussian kernel) served as features in a Cox regression of overall survival across 200 patients; the inflammatory-response pathway emerged as the strongest independent predictor (HR 1.7 per SD increase, 95 % CI 1.2 to 2.4, $p = 0.002$) after adjustment for age, stage, and treatment arm." Always state the gene-set source, the kernel, and the input data.

Practical Tips

- Use `kcdf = "Gaussian"` for log-CPM, `kcdf = "Poisson"` for raw counts; the wrong setting biases scores systematically.
- Downstream differential-pathway analysis works with a standard limma fit on the GSVA score matrix with the same design used for gene-level analysis.
- For single-cell expression, per-cell GSVA is very noisy; aggregate cells into pseudobulk first, or use signed-rank alternatives (`AUCell`, `UCell`) designed for the single-cell context.
- Always compute GSVA on a single, consistently normalised expression matrix; mixing normalisations across the input matrix produces invalid scores.
- Scores are relative within a study; do not compare raw GSVA scores across independent datasets without re-running on the combined matrix.
- For pathway exploration, the MSigDB Hallmark, KEGG, and Reactome gene sets (via `msigdbr::msigdbr()`) are the standard starting libraries.

R Packages Used

`GSVA::gsva()` for the canonical GSVA and ssGSEA implementation; `msigdbr` for MSigDB gene-set retrieval; `AUCell` and `UCell` for single-cell-friendly enrichment alternatives; `singscore` for an alternative rank-based single-sample method; `escape` for a `Seurat`-integrated GSVA workflow.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat